Author resolution. The author approves choosing exactly one VF representative of each boundary RV tag and stating constancy of the finite RV-image set along each source RV fiber. The proof now splits zero-coordinate faces before applying Corollary 2.30, retains all input tags in the graphs, and gives the zero-dimensional base case and the coordinate-chain coverage argument.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.

Lemma 2.36 ([17, Lemma 8.7])**.** *There is a definable closed set $A \subseteq \mathrm{VF}^n$ with $\dim_{\mathrm{VF}}(A) < n$ such that $f \restriction (\mathrm{VF}^n \smallsetminus A)$ is continuous.*

Per the usual terminology, this lemma says that f is continuous *almost everywhere*.

Definition 2.37. In this definition, for a nonzero tuple $x \in \mathrm{VF}^k$ we set $\mathrm{val}(x) = \min_i \mathrm{val}(x_i)$, and we set $\mathrm{val}(0) = \infty$. This minimum convention is separate from the coordinatewise valuation image used in Terminology 2.39.

For any $a \in \mathrm{VF}^n$, we say that f is *differentiable at a* if there is a linear map $\lambda : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ (of VF-vector spaces) such that, for any $\epsilon \in \Gamma$, if $b \in \mathrm{VF}^n \setminus \{0\}$ and $\mathrm{val}(b)$ is sufficiently large then

$$\mathrm{val}(f(a+b) - f(a) - \lambda(b)) - \mathrm{val}(b) > \epsilon.$$

It is straightforward to check that if such a linear function λ (represented by a matrix with entries in VF) exists then it is unique and hence is called the *derivative of f at a*.

Write $f = (f_1, \ldots, f_m)$. For $a = (a_i, a_{\tilde{i}}) \in \mathrm{VF}^n$, if the derivative of the function $f_j \restriction (\mathrm{VF} \times a_{\tilde{i}})$ at a_i exists then we call it the *ijth partial derivative of f at a*. Whenever f is differentiable at a, the (j,i)-entry of its derivative matrix is this ijth partial derivative.

Resolved referee R034 [Clarification; revised] – Valuation of a tuple and the zero increment. *Original source line 980.* Define $\mathrm{val}(b) = \min_i \mathrm{val}(b_i)$ for a nonzero tuple, with $\mathrm{val}(0) = \infty$, and require $b \neq 0$ in the difference quotient. Otherwise the display includes the undefined expression $\infty - \infty$ at $b = 0$. Later, where valuations of tuples are taken coordinatewise (Terminology 2.39), distinguish that use from this minimum convention. The input-first indexing of partial derivatives is legitimate, but please keep the corresponding matrix convention explicit.

Author resolution. The author approves the minimum coordinate valuation for differentiability, with $\mathrm{val}(0) = \infty$, and excludes the zero increment from the difference quotient. This avoids $\infty - \infty$. The input-first ijth partial-derivative convention is retained and its derivative-matrix position (j,i) is explicit. The separate boundedness convention remains coordinatewise.

We differentiate functions between arbitrary definable sets by "forgetting" the RV-coordinates. More precisely, let

$$f : \mathrm{VF}^n \times \mathrm{RV}^m \longrightarrow \mathrm{VF}^{n'} \times \mathrm{RV}^{m'}$$

be a definable function. For each fixed input RV-tuple t, generic local constancy of definable RV-valued maps [17, Lemma 8.9] implies that, outside a definable set of VF-dimension less than n, the output RV-tuple is constant on a neighborhood of each input a. If this tuple is s, then $\mathrm{dom}(f_{(t,s)})$ contains a neighborhood of a. At such a point (a,t), we define the partial derivatives of f to be those of this local VF-map $f_{(t,s)}$. If $n = n'$ and all the partial derivatives exist, the *Jacobian of f at (a,t)* is the determinant of the matrix whose (j,i)-entry is the ijth partial derivative, and is denoted by $\mathrm{Jcb}_{\mathrm{VF}} f(a,t)$.

Lemma 2.38. *Assume $n = n'$. The Jacobian of f exists and is a continuous function almost everywhere.*

Proof. First remove the definable locus where the output RV-tuple is not locally constant in the input VF-coordinates. On each resulting open fiber with both input and output RV-tuples fixed, extend the VF-map by zero outside its domain and apply [17, Corollary 9.9] to its partial derivatives. Their nonexistence and discontinuity loci are definable and have VF-dimension less than n in every such fiber. Retaining or projecting away RV-coordinates does not increase VF-dimension, so the combined exceptional locus still has VF-dimension less than n. Off this locus all partial derivatives, and hence their determinant, are continuous. In arity zero the Jacobian is the empty determinant 1. □

Resolved referee R035 [Gap] – Equal VF dimensions and generic RV constancy.
Original source line 993. State $n = n'$ in Lemma 2.38; otherwise the determinant called the Jacobian has not been defined. Moreover, the preceding existence of one full-dimensional fiber for each input t does not show that such fibers cover almost every input point. Use the standard generic local constancy of a definable map from VF^n to $\mathrm{RV}^{m'}$ first, and then apply the cited differentiability theorem on its open fibers. This provides the missing almost-everywhere coverage.

Author resolution. The author approves the square-dimension hypothesis $n = n'$ and the generic local-constancy argument for the output RV-labels. The proof now applies the cited partial-differentiability theorem on open fixed-label charts and explains the smaller-dimensional exceptional locus. The arity-zero Jacobian is the empty determinant 1.

Terminology 2.39. We say that a set $I \subseteq \Gamma_\infty^n$ is γ-*bounded*, where $\gamma \in \Gamma$, if it is contained in the box $[\gamma, \infty]^n$, and is *doubly* γ-*bounded* if it is contained in the box $[-\gamma, \gamma]^n$. More generally, let A be a subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m \times \Gamma_\infty^l$ and

$$A_\Gamma = \{(\mathrm{val}(a), \mathrm{vrv}(t), \gamma) \mid (a, t, \gamma) \in A\} \subseteq \Gamma_\infty^{n+m+l};$$

then we say that A is γ-*bounded* if A_Γ is, and so on.

Referee R036 [Clarification] – A valuation image is not a coordinate projection.
Original source line 1001. The notation A_Γ here conflicts with Notation 2.1, where it denotes projection to existing Γ-coordinates. Use a separate notation such as $\mathrm{valim}(A)$ for this coordinatewise valuation image. Also explicitly say that 'bounded' means γ-bounded for some γ, and similarly for 'doubly bounded'.

Definition 2.40 (Contractions)**.** A function $f : A \longrightarrow B$ is rv-*contractible* if there is a (necessarily unique) function $f_\downarrow : \mathrm{rv}(A) \longrightarrow \mathrm{rv}(B)$, called the rv-*contraction* of f, such that

$$(\mathrm{rv} \restriction B) \circ f = f_\downarrow \circ (\mathrm{rv} \restriction A).$$

Similarly, it is res-*contractible* (respectively, val-*contractible*) if the same holds in terms of res (respectively, val or vrv, depending on the coordinates) instead of rv.

Lemma 2.41. *Let $U \subseteq \mathrm{RV}^k$ be a doubly bounded set and $f : U \longrightarrow \mathrm{RV}$ a definable function. Then $f(U)$ is also doubly bounded. The same holds if the codomain of f is Γ.*

Proof. By induction on k, both claims are immediately reduced to showing that if $k = 1$ and $g : U \longrightarrow \Gamma$ is a definable function then $g(U)$ is doubly bounded. Then, by C-minimality, we may assume that g vrv-contracts to a function $g_\downarrow : \mathrm{vrv}(U) \longrightarrow \Gamma$. Since definable functions in Γ are piecewise $\mathbb{Q}$-linear, the range of $g_\downarrow$ must be doubly bounded. □

Referee R037 [Gap] – Uniformity in the bounded RV-image lemma.
Original source line 1017. The induction on k must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group